

Dean Fortier

- I am a robotics software engineer with broad experience in robot hardware configuration, perception, control, and robot learning.
- My experience ranges from academia and R&D consultancy to production software, middleware, and cloud solutions.
- Robotics is hands down the most rewarding field (when things are actually working 🤖)

EXPERIENCE

Microsoft, Redmond WA / Robotics Research Engineer (Contract)

OCT 2024–PRESENT

- Developed the hardware, data collection, control and assisted in training and integration for [Interactive BusyBox](#) demo - an agentic system consisting of a voice agent front end for interacting with a user, a VLM to observe the state of the BusyBox, and a VLA to control bimanual manipulation of a robot.
- First authored a paper on the [Busybox](#) - presented work at CoRL in Korea
- Assist in all phases of developing a [Rho-Alpha](#). Owned data collection and processing for multiple robots, Evaluation Rollouts, Training ablations against different SOTA models.

Optitrack, Remote / Software Engineer 2

APR 2023–OCT 2024

- Owner of ROS2 plugin for Optitrack. I created the interface and demos for ROS2 as well as adding support for ARM devices. **C++**
- With motion capture data, I was the owner for our product *MotiveBatchProcessor* which provides tools to process takes simultaneously. **C++/ C#**
- Worked in a team to develop the take processing system for next-gen Motive software. **Python**
- Responsible for CICD pipeline using github actions and self hosted servers for system testing next-gen Motive project **Python Testing/ Github CICD**

Amazon Scout, Seattle WA / Robotics SDE L4

APR 2022–NOV 2022

- Supported software for mobile robot homes (dispensers) consisting of: large data offload from robots to S3, monitoring docked robots, software updates to dispensers and robots.
 - **Track Operational Misses** - in seeing data loss due to human error, I created a dashboard, AWS QuickSight, that tracks data dropout so we can better support operations teams **Python/ Linux/ AWS**
 - **Improved Robot Heartbeat** - In order to better track if a robot has docked I introduced a heartbeat on each robot compute to the dispenser's server using AWS Lambda and CloudWatch **Python/ Linux/ AWS**

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EDUCATION

University of Florida

Electrical Engineering

FALL 2014 - SPRING 2019

Data Pipelines with TensorFlow Data Services - 2024

Udacity Deep Learning Nanodegree - 2021

Udacity Robotics Software Engineer Nanodegree - 2020

SKILLS

C++	Python
Computer Vision	PyTorch
Sensor Fusion	ROS
Motion Capture	Linux
C#/Managed C++	Docker
Azure	AWS

HOBBIES

Music/ Festivals
Photography
Biking
Sitting next to Camp Fires

Fresh Consulting, Bellevue WA / Robotics Engineer

JUN 2019–APR 2022

- Worked as a robotics engineer in teams ranging from 2 to 10:
 - **Miso Robotics** - Consulted design solutions for Miso and Created Software adaption layer to support a Yaskawa 6DOF robot arm for Flippy 2 robot system for automated cooking. **C++/ Python/ ROS**
 - **Microsoft** - Developed a puck tracking to operate 200 Hz and gantry controller for an autonomous air hockey table **C++/ Computer Vision/ CUDA/ Control Theory**
 - **Ventec** - Developed test automation for ventilators capable of testing up to 24 ventilators simultaneously **Python/ RTOS Concepts/ SQL**
 - **John Deere** - Designed and implemented 6-camera calibration method used for birds-eye view perspective at 30 FPS **C++/ Computer Vision/ Networking**
 - **United Rentals** - Added Camera localization to the overall localization stack, Deployed robots and Trained warehouse workers to use our robot. **ROS/ Networking/ Computer Vision**
 - **Hyundai** - Researched using Reinforcement Learning control in a wheel-quadruped robot and implemented a simulated prototype. **ROS2/ Simulation/ RL/ Docker**
 - **SBQuantum** - Deployed cost-effective outdoor AGV capable of surveying 1 km² using GPS, IMU, and wheel odometry **ROS/ Sensor Fusion/ Docker**

FOCUS Lab University of Florida / Research Assistant

OCT 2018– FEB 2019

- Designed a test setup and Created an Abstraction layer to control MEMs Mirror reflecting a laser beam **Python**
- Mirror and Camera were co-located on an optical table. Camera server node would request point scan at a particular pixel coordinate via UDP. Given Camera intrinsics and sensor size, the client (mirror) node would direct Mirror angle at desired pixel location **Computer Vision/ C++**

Magic Leap, Fort Lauderdale FL / Electrical Hardware Intern

MAY 2018 – AUG 2018

- Designed test procedure to precisely find Magnetic Sensitivity of EM coils for 6DoF tracking **Python/ LabView/ Oscilloscope/ Signal Generator/ Amp**
- Verified DisplayPort 1.4 physical compliance to test cables' signal integrity **LabView/ Oscilloscope**